📘 MODULE 5: SOURCES + SOURCETYPES

5.1 Concept

When data arrives, Splunk tags it with host, source, and sourcetype. The **sourcetype** tells Splunk how to break lines into events, find the timestamp, and extract fields.

5.2 Automatic vs manual assignment

Field	Automatic	Manual
host	Name of the machine that sent it	host = in inputs.conf
source	File path or input name	Rarely overridden
sourcetype	Splunk guesses from the file name/content	sourcetype = in inputs.conf or Add Data wizard

Best practice: set the sourcetype **explicitly** at the input. Guessing causes wrong parsing.

Example inputs.conf stanza (/opt/splunk/etc/system/local/inputs.conf):

ini

[monitor:///var/log/auth.log]

index = linux_auth

sourcetype = linux_secure

disabled = 0

- monitor:// watches a file or folder (note the three slashes: :// plus the leading / of the path).
- index where to store · sourcetype the format · disabled = 0 means enabled.
- Restart or reload for changes. Verify with a search.

5.3 Custom sourcetypes

Create one when no standard sourcetype fits. In props.conf:

ini

[my_app_logs]

SHOULD_LINEMERGE = false

TIME_PREFIX = ^

TIME_FORMAT = %Y-%m-%d %H:%M:%S

MAX_TIMESTAMP_LOOKAHEAD = 19

- [my_app_logs] is the sourcetype name.
- SHOULD_LINEMERGE = false: one line = one event.
- TIME_PREFIX/TIME_FORMAT: where and how the timestamp looks.
- We cover props.conf deeply later. For now just recognize it.

Naming convention: vendor:product:type (e.g. cisco:asa). Standard names come from add-ons.

5.4 Why these searches differ

Query A

spl

index=* source="/var/log/auth.log"

Matches events from that **exact file path**. A different file with the same format would not match.

Query B

spl

index=* sourcetype=linux_secure

Matches **all data of that format**, from any file, any host. This usually gives more results than A.

Common mistakes: a wrong sourcetype name (case-sensitive values matter), or a source path with a typo. Wildcards work: source="*auth*".
Interview explanation: “Source filters by origin, sourcetype by data format. I use sourcetype when I want all data of one kind regardless of where it came from.”

5.5 Why sourcetype matters to a SOC

Correct sourcetype → correct timestamp and fields → CIM mapping works → detections and dashboards work. Wrong sourcetype is a top cause of “fields are missing.”

5.6 Hands-on

1. Run index=* | stats count by sourcetype. What sourcetypes do you have?
2. Run index=* | stats count by source. Compare.
3. Pick a sourcetype and note how many sources feed it.

4. Update your monitor stanza as in 5.2, restart, and check the sourcetype changed (new events only; old events keep the old sourcetype).

5.7 Interview Q&A

- **Source vs sourcetype?** Source is origin, sourcetype is format and controls parsing.
- **How is sourcetype assigned?** Automatically by Splunk, or manually in inputs.conf/props.conf/Add Data. Manual is best practice.
- **Wrong sourcetype symptom?** Broken event splitting, bad timestamps, missing fields.
- **Does changing sourcetype fix old events?** No. Indexed data isn't rewritten; only new events are affected.

5.8 Revision

host=machine · source=origin · sourcetype=format · set sourcetype explicitly · inputs.conf assigns, props.conf defines behavior · old data isn't rewritten.